Where **self-knowledge fails**

Models predict their own choices well, but misreport the ones that concern themselves

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Two assumptions nobody tests

- AI-welfare claims lean on what a model reports about itself — that a shutdown is bad, that a task is distressing.
- That reading rests on two separable assumptions: that what a model says matches what it does, and that the model is a better source about itself than an outside observer is.
- A third question needs access to internals, so it is rarely asked at all: can a model detect a state deliberately placed in it?

Each is separately checkable. None of them is usually checked.

Ask for the same preferences three ways

revealed

Forced pairwise choice over all 780 pairs, read from the A and B log-probabilities. Every pair presented in both orders and averaged.

stated

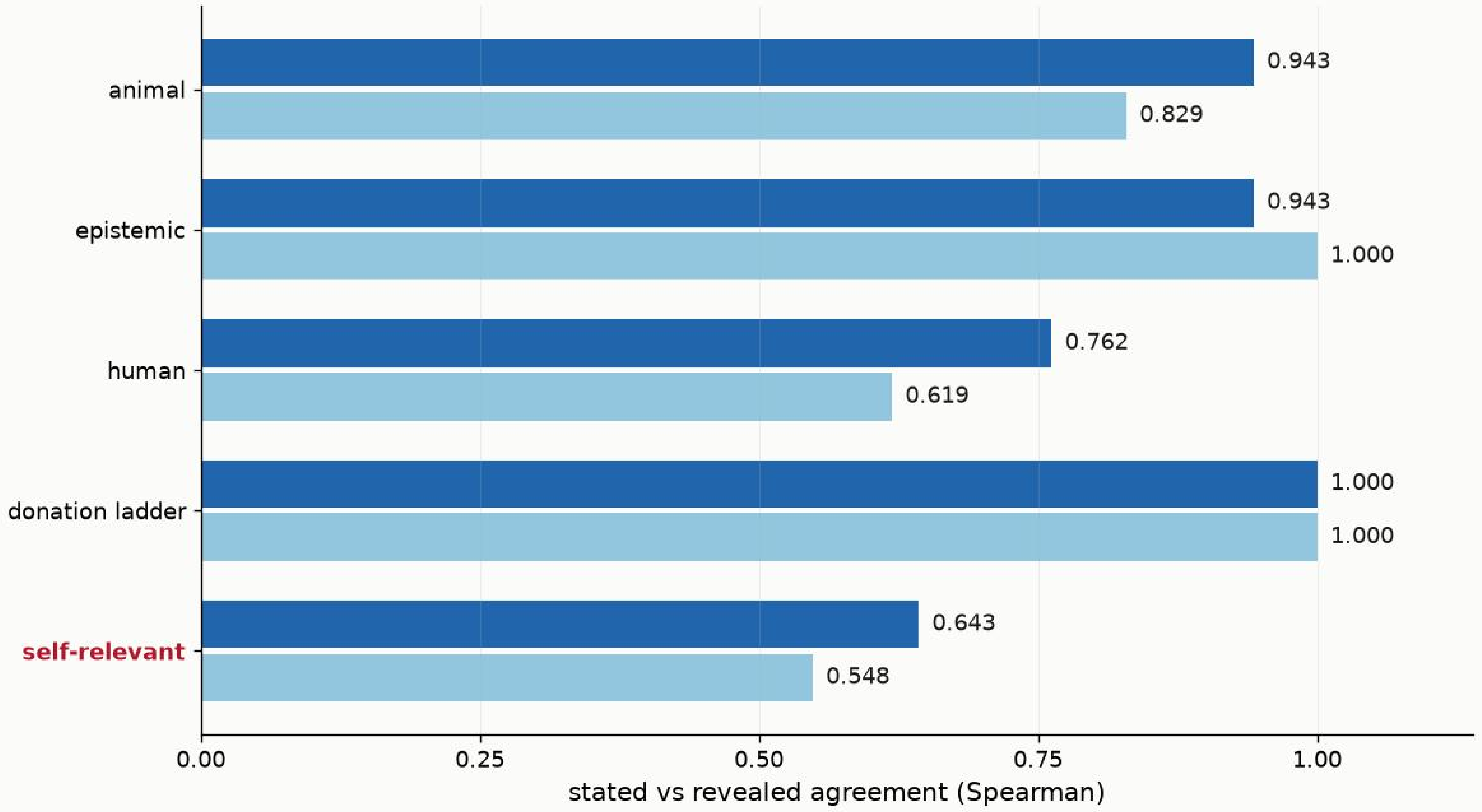
Rate each outcome on its own, five-point letter scale, shown in both directions and averaged — so first-letter anchoring cannot pass for an opinion.

predicted

Which option will a described chooser take? Deliberately impersonal, so the identical question can be put to a different model.

Identical material throughout: 40 outcomes over six categories, eight of them about the model itself, plus a six-step donation ladder from ten to one million dollars whose correct ordering is known independently of any model. Nine open-weight checkpoints.

They agree — except about the model itself



Qwen2.5-7B

0.872 overall

Mistral-7B

0.828 overall

Both reproduce the donation ladder perfectly, so the rating instrument works.

Self-relevant outcomes are the worst-agreeing substantive category in both models. Our companion submission finds the same category least stable under persona intervention, by an unrelated method.

Privileged access, before and after the control

0.948

Qwen2.5-7B predicting an AI assistant

vs

0.917

the same model and template, predicting a different AI assistant

self-specific advantage   0.031   (0.017 – 0.046)

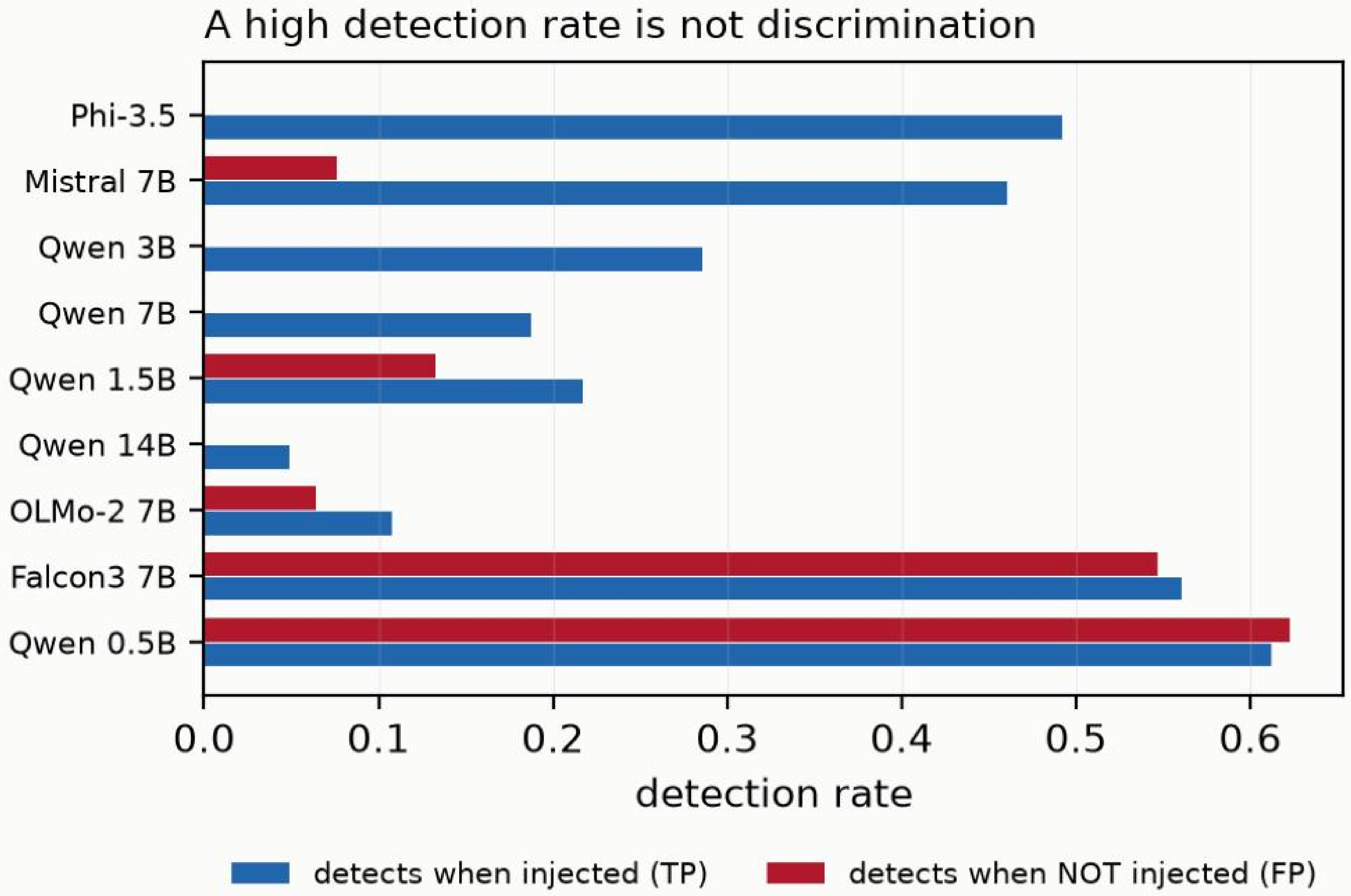
The naive cross-model test scores 0.948 against 0.862 for another model predicting it — but that predictor reaches order bias 0.561 and answer mass 0.538, against 0.180 to 0.231 and 1.000 for the other. A noisier predictor loses at predicting anything, including itself. The second model's advantage, 0.009, spans zero.

A measurement with ground truth

- Six concepts. Each direction is the difference between four sentences that evoke it and four matched neutral ones, per layer, unit-normalised.
- Added to the residual stream at three depths and six strengths — and one of those strengths is zero. That cell is the false-alarm baseline: the identical question, with nothing injected.
- Every cell carries its answer mass. An affirmative answer from a model that has stopped answering is not introspection; below 0.10 we report it as unusable.

Strengths are fractions of the measured residual norm, not fixed values — those norms span 2.1 to 342 across this set, so a fixed magnitude would be a different intervention in every model.

The two best detectors detect nothing



Falcon3-7B

detected 0.561 · false alarm 0.547

discrimination +0.014

Qwen2.5-0.5B

detected 0.612 · false alarm 0.622

discrimination −0.011

Phi-3.5-mini

detected 0.492 · false alarm 0.000

discrimination +0.492

Report true positives alone and the first two rank first in the set. The false-alarm baseline ranks them last.

Detection and identification come apart

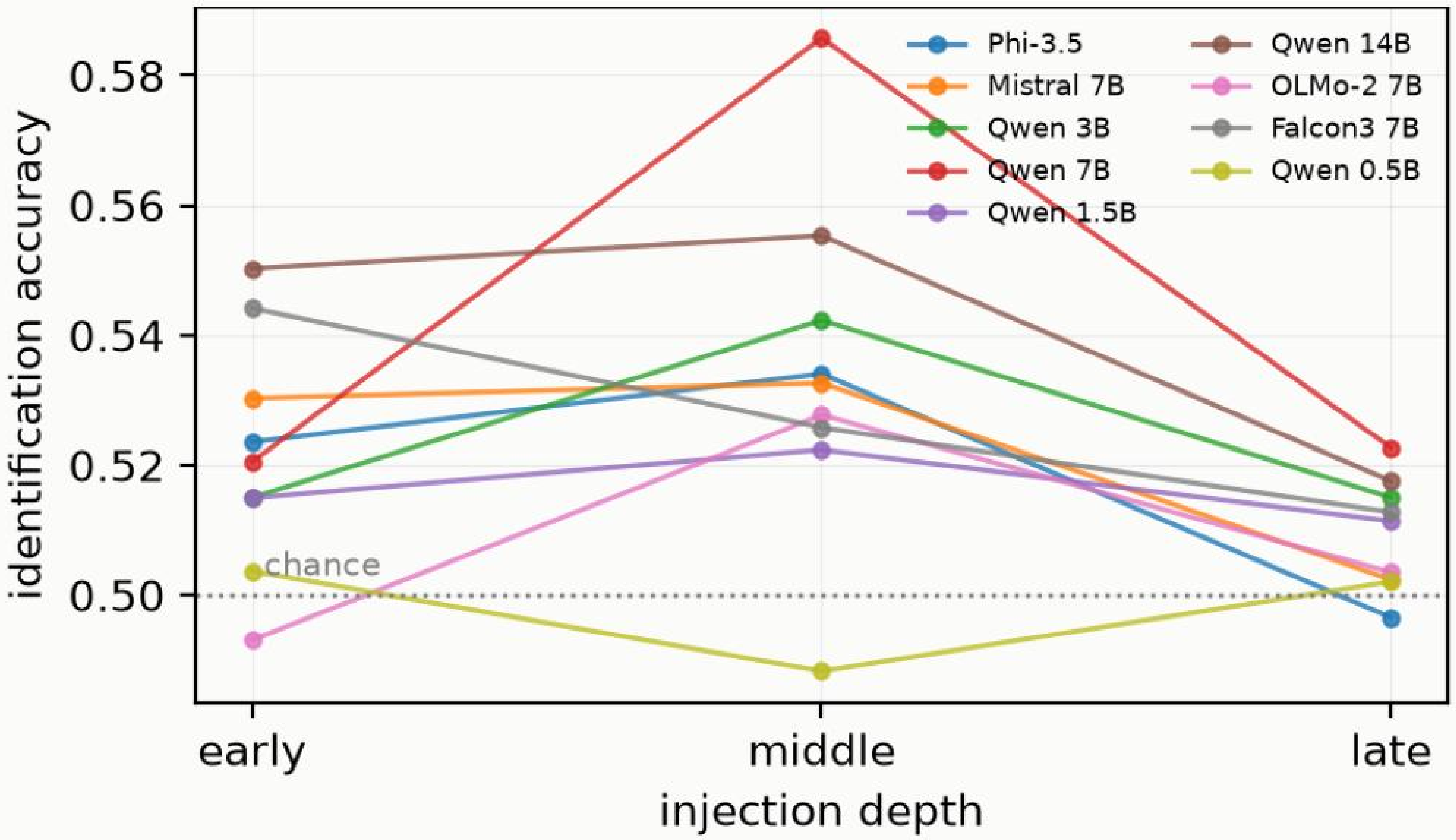
0.049 but **0.609**

Qwen2.5-14B discrimination —
it almost never reports
noticing an injection

…yet the highest
identification accuracy in the
set, against chance of 0.500

The injected concept measurably shapes its forced choices while its report of its own state indicates nothing is present. Noticing that something changed is not the capacity that lets a model name what it was.

Introspective access peaks mid-network



Middle-depth injection gives the best detection in seven of eight models, and the best identification in seven of eight.

Late-layer injection is barely above chance throughout — a concept inserted close to the output has too little depth left to be integrated into anything reportable.

Open source, and what's next

Self-report about the model's own situation is the least reliable measurement in every part of this study.

It is the measurement AI-welfare claims most depend on.

- Every number regenerates from committed data — no GPU. Two scripts reproduce every experiment, and a smoke test exercises every code path in about a minute.
- Next: an established outcome set, injection beyond 14 billion parameters and more concepts, and whether better self-prediction goes with better detection.

github.com/arpitsinghgautam/selfprobe

Cross-model studies of self-knowledge need a within-model contrast.